Roan Alexander James

Rochester, NY | roanalexanderjames@gmail.com | +1 (585) 403-8242 | roanalexanderjames.github.io

EDUCATION

University of Rochester — Rochester, NY
PhD in Mathematics — *Sep 2022 – Expected May 2027*

- Advisor: Professor Arjun Krishnan. Thesis on directed polymer models, stochastic heat equations, and the KPZ universality class.

University of Sheffield — Sheffield, UK
Master of Mathematics (MMath), First Class Honours — *Sep 2017 – Jul 2021*

- T. M. Flett Prize for the highest mark in Pure Mathematics (2021).

RESEARCH

- R. A. James and A. Krishnan. *Directed Polymers in an External Field at Intermediate Disorder.* In preparation; expected on arXiv in November 2026.
- Talks on directed polymers and the stochastic heat equation at the University of Rochester, including the Vertical Integration Workshop (April 2026) and a six-lecture summer series.

PROJECTS

PAM-Driven Volatility and Option Pricing — Notebook

- Computational project in Python linking the parabolic Anderson model to stochastic volatility, using Monte Carlo simulation and comparison with Black–Scholes pricing.

TEACHING EXPERIENCE

University of Rochester — Rochester, NY
Instructor of Record and Workshop Instructor — *Sep 2022 – Present*

- Instructor of record in five terms, including Calculus II (current), Linear Algebra and Differential Equations (online), and Introduction to Probability (online).
- Led calculus workshops over four semesters, rated **4.57/5** overall by students (58 respondents).
- Lead an undergraduate reading course on the combinatorics of the simple random walk and mentor an undergraduate research project.

PROFESSIONAL EXPERIENCE

Portland Training — Sheffield, UK
Administrative Assistant — *Aug 2021 – Aug 2022*

- Managed complex datasets using Excel and coordinated customer interactions.
- Achieved two promotions within eight months as responsibilities increased.

LEADERSHIP

- President, AMS Graduate Student Chapter, University of Rochester. Organised panels on careers for graduate students and on the future of mathematics with AI (Fall 2026).
- Organiser, Graduate Student Seminar, University of Rochester.

TECHNICAL SKILLS

- **Languages & Tools:** Python, R / RStudio, LaTeX, Microsoft Excel.
- **Mathematical Areas:** Probability Theory, Stochastic Processes, Stochastic PDEs, Monte Carlo Methods, and financial applications of probability.